FNIH Partnership Concept: A Pre-Competitive Perioperative RAS Consortium in Resectable KRAS-Mutated PDAC

Public side	+	**Private side**
An independent scientist holding a published protocol, an IND, and an audited simulation package, all openly deposited.		A qualified academic pancreatic surgery and medical oncology program, plus investigational drug access under a cross-reference.

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Submitted as an independent scientist under the funding approach set out in Science: A New Golden Age *(July 2026). Not medical or regulatory advice, and not endorsed by the FDA, NIH, HHS, FNIH, an IRB, ICH, or any sponsor. UC San Diego is named as the intended partner of choice and is not a sponsor, site, or endorser.*

1 The Partnership Model the Report Names

Science: A New Golden Age holds up the Foundation for the NIH and the Accelerating Medicines Partnership as its model for public-private structure, noting that the Alzheimer's partnership, one of twelve disease-focused AMPs, experimentally validated twenty candidate drug targets, depicted for the current application in Table 1 [1]. In the same chapter the report observes that American discoveries are increasingly tested abroad because per-patient trial costs run far higher here, and its annex asks agencies to prioritize non-federal cost share so that federal dollars draw in private investment rather than standing alone [2].

This concept applies that structure to a narrow, pre-competitive question in resectable pancreatic cancer: what happens pharmacologically when a RAS(ON) multi-selective inhibitor is given around a curative-intent resection. No single party can answer it. The drug developer holds the compound, an academic center holds the patients and the operative expertise, and the question that matters, what the resection specimen shows, is a public good that no company will fund on its own because it does not differentiate a product.

AMP feature	Instance proposed here
A pre-competitive question no party owns	Perioperative RAS pathway pharmacodynamics measured in resected human PDAC tissue
Shared governance with public reporting	Protocol, analysis plan, and negative results deposited before database lock
Non-federal cost share drawing in private investment	Investigational drug supply and site infrastructure contributed rather than purchased
Data released to the field, not to a sponsor	Specimen-level pharmacodynamic dataset released under an open license

Table 1: Four defining features of the Accelerating Medicines Partnership model, and the specific instance each takes in a perioperative RAS consortium built around a resectable pancreatic cancer trial.

2 Who Contributes What

The order in which the three parties commit matters more than the totals, illustrated in Figures 1 and 2. A drug cross-reference that arrives after the IRB submission forces a resubmission; a site agreement that arrives after the drug supply wastes the supply window.

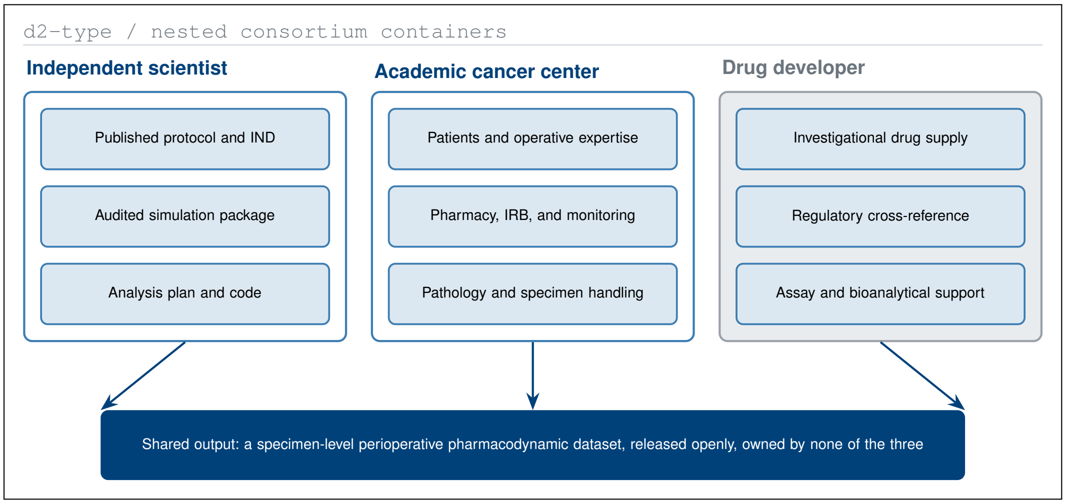


Figure 1: The three contributing parties and the single shared output. Each container holds only what that party can supply, and the output belongs to none of them, which is what makes the question pre-competitive.

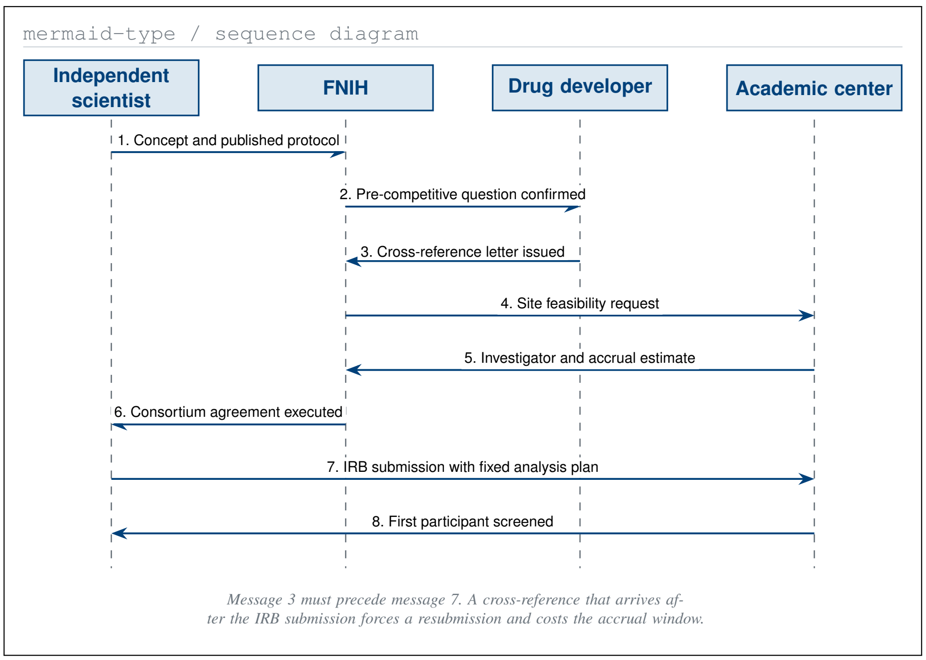


Figure 2: The order in which the four parties must act. The dependency that governs the schedule is message 3 preceding message 7, not the size of any one contribution.

3 Pharmacologic Rationale and the Prior Record

Daraxonrasib (RMC-6236) is a RAS(ON) multi-selective inhibitor. In May 2026, RASolute 302 reported median overall survival of **13.2 months** against **6.6 months** for chemotherapy in the RAS G12 population of previously treated metastatic pancreatic cancer, seen in Table 2 [3]. Nine months earlier, a ten-arm quantitative systems pharmacology simulation with 250 ordinary differential equations per patient had returned **12.8** against **5.4 months** for a daraxonrasib combination [4].

The proximity of those two results is a chronology observation and a hypothesis-supporting one. It is **not** a validation claim. Three differences are material and are stated wherever the comparison appears: sample size, 1000 simulated against 241 enrolled; intervention, a daraxonrasib combination against daraxonrasib as a single agent; and molecular selection, KRAS G12C against a primarily G12D and G12V population.

Source and date	Test arm	Control	Reading
QSP simulation, Aug 2025	12.8 mo mOS	5.4 mo mOS	Best arm HR 0.25; 6 of 7 archetypes below HR 1.00
Digital twin, Oct 2025	12.1 mo mOS	not applicable	Best mPFS HR 0.31; credibility 81.9 under ASME V&V 40
RASolute 302, May 2026	13.2 mo mOS	6.6 mo mOS	Independent readout, RAS G12 metastatic population

Table 2. The simulated and observed results side by side. The 2025 simulated 2.4-fold survival ratio and the 2026 observed 2.0-fold ratio are reported as a chronology and not as a validation of the model.

None of that evidence speaks to the resectable setting, which is precisely why the consortium question is worth asking. The metastatic readout tells the field that the pathway is druggable in humans; it does not say what a curative-intent resection specimen looks like after perioperative exposure.

4 The Trial as the Consortium's Instrument

The clinical vehicle is a Phase 1, first-in-human, open-label, single-arm study in resectable or borderline-resectable KRAS G12-mutated PDAC: perioperative daraxonrasib with 3+3 dose escalation, up to 18 treated participants, and a staged eight-arm robotic pancreaticoduodenectomy with an on-premises language model confined to advisory output [5, 6]. The operation is not incidental to the pharmacology. It is the only setting in which paired pre-treatment and on-treatment human tumour tissue can be obtained under controlled timing.

Consortium measurement	Timing relative to resection	Why it is pre-competitive
Plasma pharmacokinetics	Days −14, −1, and +1	Establishes exposure in a surgical population no sponsor has characterised
Tumour pathway pharmacodynamics	Resection specimen, day 0	Cannot be obtained in the metastatic setting at all
ctDNA clearance kinetics	Days −14, +30, +90	Defines a shared surrogate the whole field can reuse
Pathologic response grade	Resection specimen, day 0	Anchors every later perioperative RAS study to one scale

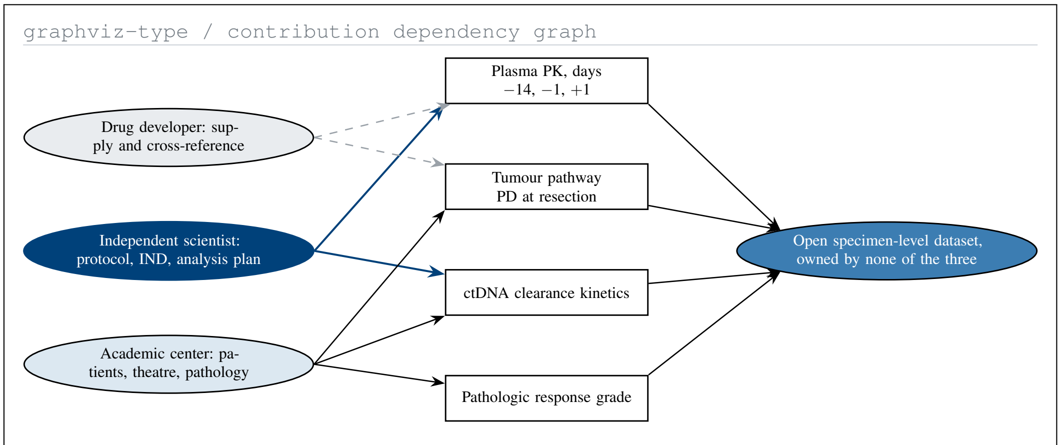
Table 3. Four measurements the consortium exists to make, when each is taken relative to the resection, and the reason none of them differentiates a commercial product.

Positioning note. *First in human refers to the integrated surgical and advisory workflow, not to daraxonrasib, which is investigational and already in Phase 3 evaluation. The robotic configuration is specified by manufacturer, model, cleared configuration, arm count, and software version at the site agreement, and no drug supply or letter of authorization is in place with the agent's developer.*

5 Cost Share, Leverage, and the Partner Site

The federal or foundation ask is **$700,000 per year for five years**, $3,500,000 total [7]. The consortium's contributed value, drug supply, site infrastructure, pathology, and bioanalytical support, is not requested in cash and is the non-federal share the annex asks agencies to prioritise, based on Figure 3 [2].

Moores Cancer Center is the intended partner of choice for the academic side, approached at the feasibility stage only. UC San Diego is not a partner, sponsor, site, or endorser, because no written authorization exists.



Two of the four measurements have two independent upstream contributors and survive the loss of one party. Tumour pathway pharmacodynamics does not, which is why the drug cross-reference is the first agreement to close.

Figure 3. Which contribution unblocks which measurement. Only tumour pathway pharmacodynamics depends on a single party on each side, which sets the order in which the consortium agreements must close.

Scope of Claims

Claimed: a published protocol and IND, an audited simulation package, and a consortium design whose single shared output is owned by none of the three contributing parties. Not claimed: that the simulations validate RASolute 302, that the advisory system holds any regulatory clearance, that anything here has been used in a human, or that UC San Diego has agreed to anything. The trial does not exist until a site, an IRB, an active IND, monitoring, insurance, and a sponsor of record are all in place, visualized in Table 3.

Availability and Conflicts

Every work cited to a DOI is openly deposited; code, verification notebooks, and figure sources are in the repository [8]. The applicant holds no equity in Revolution Medicines or any robotic surgery vendor, and no sponsor, contract research organization, or medical society reviewed this document.

References

[1] Michael Kratsios. Science: A New Golden Age. Report to the President of the United States, July 2026. Letter of Transmittal, July 21, 2026. URL: https://www.whitehouse.gov/wp-content/uploads/2026/07/Science-A-New-Golden-Age.pdf.

[2] Michael Kratsios and Russell T. Vought. Fiscal Year 2028 Administration Research and Development Budget Priorities. Joint OSTP and OMB memorandum NSTM-5 / M-26-16, July 2026. Annex to Science: A New Golden Age; names the six national missions. URL: https://www.whitehouse.gov/wp-content/uploads/2026/07/White-House-Fiscal-Year-2028-RD-Priorities-Memorandum.pdf.

[3] Eileen M. O'Reilly et al. Daraxonrasib or Chemotherapy in Previously Treated Metastatic Pancreatic Cancer. *New England Journal of Medicine*, May 2026. doi:10.1056/NEJMoa2605555.

[4] Kevin Kawchak. QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation: From Protocol to Prediction, Code, VVUQ, and Playbook. Zenodo, August 2025. doi:10.5281/zenodo.17001137.

[5] Kevin Kawchak. On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma. Zenodo, June 2026. doi:10.5281/zenodo.20780121.

[6] Kevin Kawchak. Investigational New Drug Application: Daraxonrasib, Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC. Zenodo, July 2026. doi:10.5281/zenodo.21097442.

[7] Kevin Kawchak. Clinical Trial Funding Application v2.0, RFA-RM-27-001. Zenodo, July 2026. doi:10.5281/zenodo.21317266.

[8] Kevin Kawchak. physical-ai-oncology-trials, repository v4.4.0. GitHub, 2026. URL: https://github.com/kevinkawchak/physical-ai-oncology-trials.